

II.9

ANOTHER SCIENTIFIC REVOLUTION

The occult sciences in theory and experimentalist practice

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The Scientific Revolution, a problematic 20th-century label that nevertheless faithfully conveys the ethos of many 16th- to 19th-century sources, is central to our narrative of the emergence of Western modernity. That revolution was multidisciplinary and progressive: first in astronomy and physics (Copernicus to Newton), then in chemistry and biology (Boyle to Darwin). In all fields, it was defined by a new empiricist and experimentalist program, whereby even the authority of Aristotle could now be set aside, albeit respectfully, when it contradicted direct observation (Galileo and Bacon). And while its protagonists were not revolutionary in the counterintuitive modern sense of the term, instead championing their empirical science as the revival (“the Renaissance”) and extension of ancient, perennial wisdom, most shared a millenarian identity as the self-styled architects of a new age (Shapin 1996; McKnight 1998).

To that end, many of our Heroes of Science were also deeply invested in the occult sciences, from astrology and alchemy to geomancy and magic, as integral to their natural philosophy. Yet this simple fact long horrified modern historians, who until recently sought to winnow the chaff of such “medieval superstition” from the grain of True Science, engine of the Rise of the West. Such a heresiographical and teleological approach is no longer considered legitimate by historians of science. We now emphasize rather the messy, meandering, orthodoxizing and often abortive nature of “scientific progress,” present and past, which was historically achieved by diehard occultists (like Newton and Bacon) as often as not (see, e.g., Yates 1964; Langman 2010; Newman 2019).

The same principle applies to the contemporary Islamicate and especially Persianate world, culturally just as “Western” as the Latinate world in its dual focus on Hellenic philosophy and natural science and Abrahamic monotheism, whose imperial states (Mughal, Safavid, Uzbek, Ottoman) encompassed a full third of the human race (Melvin-Koushki 2018a, 2018b). That Islamdom was far wealthier and more populous and cosmopolitan than Christendom throughout the early modern period, moreover, meant that the sciences – especially the occult sciences – flourished there to an even greater degree. Astrology, letterism, geomancy, oneiromancy and physiognomy in particular enjoyed heavy royal patronage (see, e.g., Şen 2016; Lelić 2017; Melvin-Koushki 2018d). And no wonder: for these and their allied divinatory and magical disciplines promised control, especially *imperial* control, of minds and bodies, time and space. First you must predict the future; then you can change it.

The explosion in patronage and popularity of such disciplines in courtly and scholarly circles around the turn of the 9th/15th century is reflected by the sudden emergence of a new Arabo-Persian genre: the imperial grimoire (Melvin-Koushki 2018c). Unparalleled in the Latinate world, where political and economic units were smaller, these occult-scientific manuals are explicitly modeled on the vastly popular 4th/10th-century Pseudo-Aristotelian *Secret of Secrets* (*Sirr al-asrār*, Lat. *Secretum secretorum*), wherein the Greek sage teaches his protégé Alexander the Great that world empire can only be achieved and maintained by means of astrology, alchemy, physiognomy and magic. Similarly, some early modern Muslim sages sought to turn their royal patrons into mages by writing highly accessible manuals on these and other occult sciences – and sometimes succeeded in this project. In the Ottoman case, indeed, one such manual, ‘Abd al-Raḥmān al-Biṣṭāmī’s (d. 858/1454) *Key to the Comprehensive Prognosticon* (*Miftāḥ al-jāfr al-jāmi’*), even came to function in the 10th/16th century as a *dynastic handbook*, whereby the political fate of the House of ‘Oṣmān could be effectively managed over the generations, even to the Last Hour itself (Fleischer 2009). Astrology, too, was so thoroughly institutionalized at the Ottoman court that the office of court astrologer only ceased with the fall of the Ottoman Empire in 1922, almost four and a half centuries since its institution (Şen 2016). Competing Islamicate empires used similar strategies to jockey for precedence. Early modern Islamicate imperialism, in short, was pinned to this specifically occult-scientific model of the Alexandrian Empire, synthesized with Mongol, Persian and Islamic precedent, and hence often pursued as an occult science in its own right (Melvin-Koushki 2018a, 2018c).

But the occult sciences were not of merely political, economic and military interest. Just as in West Europe, they were held up by some of the greatest Muslim philosophers and saints of the early modern era – who, again, often functioned as court mages and architects of Platonic Empire – as the basis for what we might call a *scientific revolution*. For all their perennialist respect for the ancients, these ambitious thinkers consciously achieved an unprecedented and fundamental break with Hellenic antiquity, while avidly using its tools. That is, they consciously engineered a paradigm shift, in Kuhnian terms. And they too thought of themselves as *modern* (Melvin-Koushki 2018b).

This new cosmological imaginary was born in the 6th/12th-century western Mediterranean, a beneficiary of climate change; began to crystallize in 7th/13th-century Mamluk Egypt and Ilkhanid Iran, where it was heavily supported by new forms of post-caliphal empire; then was given wings by the 8th/14th-century apocalypse that was the Black Death. Like other crisis-born cosmologies before and since, it was remarkably long-lived and would enjoy hegemony throughout the Islamicate world down to the colonial period, at all levels of society. The sheer profusion of relevant surviving manuscripts and artifacts aside, the surest index of the ubiquity of occult science in post-Mongol Islamicate societies is the fact that market inspectors (*muḥtasib*) had to be appointed to regulate this booming industry (Mauder 2019). Unlike our own, however, the cosmology that underpinned it was emanationist and analogist, which is to say, Neoplatonic – and often just as millenarian as Newtonian and Baconian science. As Noah Gardiner recently put it:

In this new regnant vision of the cosmos, the stars, planets, plants, minerals, bodily humors, colors, scents, letters of the alphabet, numbers, nations, and countless other entities and accidents of creation were enmeshed in a web of sympathetic or antipathetic relationships that vibrated along the ‘great chain of being’ between the godhead and the manifest world.

(Gardiner 2019, 557)

To a majority of early modern Muslim thinkers, occult sciences thus were simply those natural and mathematical sciences that could best penetrate and manipulate the structure of the cosmos to theoretical and practical advantage – and so help to divinize man, to achieve Neoplatonic *theosis*.

Most crucially for the history of science, this paradigm shift was also a specifically Neopythagorean one: the world would now be read in exclusively *mathematical-linguistic* – which is to say, *talismanic* – terms. This is a major source of the Two Books doctrine, peculiar to the bibliomaniac West, whereby the natural philosopher (Muslim, Jewish or Christian) must treat nature and scripture as twin expressions of the same divine revelatory mind. Henceforth the world was to be read, and magically rewritten, as a *mathematical text* (Håkansson 2001; Harrison 2001; Howell 2001; Killeen and Forshaw 2007).

Now this *mathesis*, of course, has been hailed by historians as the very heart of the Scientific Revolution. After all, did not Galileo declare that “the book of nature is written in the language of mathematics”? And were not Copernicus and Kepler ardent Pythagoreans (Hallyn 1990)? (While not consciously Pythagorean themselves, modern physicists still embrace both men as exemplars; and the new physics of information is curiously Neopythagorean in tenor [Wheeler 1990, Vlatko 2010, Melvin-Koushki 2016].) The overtly occult-scientific program that drove this shift, which fundamentally reorganized the human relationship to the cosmos, and which gained traction in the Islamicate world well before the Christianate, was thus no less important to Western intellectual and cultural history than the Copernican shift of a few centuries later – and even, recent scholarship suggests, helped make the latter possible (Saliba 2007[CB];¹ Ragep 2007; Westman 2011; Morrison 2014; Feldhay and Ragep 2017; Melvin-Koushki 2017). True, Muslim scholars remained formally committed to geocentric theory throughout the early modern period, out of deference to Aristotle and Ptolemy, or simply ignored the question (at least as far as we know). But the boom in Neoplatonic and especially Neopythagorean philosophy, the natural epistemological basis for heliocentrism since antiquity, emboldened reflective Westerners to finally break with Aristotle’s physics. It must be emphasized that the Copernican theory was seen as essentially Neopythagorean by its proponents and opponents alike, including Copernicus himself. The same epistemology also helped inspire them to begin to read the Book of Nature systematically as a second scripture. And this reading flourished as the Platonopolis flourished: for science, occult or otherwise, both drives and is driven by empire.

II.9.1 Putting the “occult” in the occult sciences

Why was this earlier, Islamic Scientific Revolution specifically Occult? What was it about the occult sciences that made them so compelling to a majority of early modern Muslim scholars and rulers, not to mention their Jewish and Christian contemporaries? Why is their surge in popularity our best index for the beginning of Western *mathesis*, and hence modernity itself? It was not simply the status of astrology or alchemy as inductive natural sciences, according to the venerable Avicennan classification – although this feature alone would have ensured their persistence in post-Avicennan scholarly culture. Nor was it merely their obvious utility as technologies of empire; astrology, for example, had already been a vaunted imperial discipline in the Near East for millennia. Rather, as sciences concerned with extrapolating from visible data to invisible, from *zāhir* to *bāṭin*, they acquired an epistemological mobility not granted to any other rational (*‘aqlī*) or traditional (*naqlī*), that is, Greek or Islamic, sciences.

The occult sciences, in short, became the only sciences in the Islamicate tradition to transcend the strict epistemic and generic divide between the natural and mathematical sciences, on one hand, and the religious and literary sciences, on the other. By the 10th/16th century,

lettrism (*ilm al-ḥurūf*) and alchemy (*ilm al-kīmīyā*) in particular were sciences classified and practiced as natural, mathematical and religious *simultaneously*. As such, they were embraced by philosophers as an especially effective means of reconciling the otherwise irreconcilable Neoplatonic emanationist and Quranic creationist accounts of the origin and destiny of humanity and the cosmos, thereby marrying Greek philosophy to its Islamic actualization (Melvin-Koushki 2019). So compelling was this concordist project, in fact, that European Renaissance humanists, from Giovanni Pico della Mirandola (d. 1494) onward, began to seize upon *kabbalah*, lettrism's coeval Hebrew twin, to much the same end. Thenceforth lettrism and kabbalah would be the primary driver of and vehicle for Neopythagoreanism throughout the early modern West (Melvin-Koushki 2016).

This transcendent quality is captured in the very term by which these sciences were routinely identified as an epistemologically coherent group, both in Arabic and in Latin: the *occult* sciences. As in modern English usage, *occult* (Ar. *khafī*, Lat. *occultus*), simply meaning “hidden,” originally conveyed an air of mystery; the sciences so designated were often taught and practiced by individuals or within closed groups – “esotericist reading communities” – who jealously guarded them against profanation by non-initiates (Gardiner 2014). Unlike the current English term, however, *occult* was not meant as a slur, synonymous with *spooky* or, better, *paranormal*, but rather as an assertion of social and spiritual superiority and exclusivity – while also embracing the frequent, wonder-inducing *weirdness* of human experience more generally. As such, it was above all a simultaneously epistemological and technological designation: the sciences so named are those that allow one to penetrate and harness the unseen, subtler, weirder, more wondrous levels of existence beyond ordinary human ken; hence the common appellations *al-ʿulūm al-laṭīfa*, the subtle sciences, and *ʿulūm al-ghayb*, the sciences of the unseen. That we now prefer instead the neologistic prefixes *dark*, *sub-*, *super-* and *para-* (as in *dark matter*, the *subconscious*, the *supernatural*, *parapsychology*) to designate the same category merely testifies to our continued belief in the magical power of naming!

Because these sciences required both rigorous intellectual training and spiritual purity, however, few scholars achieved full mastery of one or more of them. But some of those who did became renowned as true Renaissance men, celebrated in biographical dictionaries and chronicles as the greatest scholars of their age. By the early modern period, philosophers famed as both jurists and occult scientists, like the Safavid sage-mage Shaykh Bahāʾī (d.1030/1621), could thus be officially eulogized in Persian as being ‘possessed of eternal forms of knowledge (*ʿarīf-i maʿarīf-i azālī*), a master of sciences occult and manifest (*vāqīf-i ʿulūm-i khafī u jāhī*)’ (Iskandar Beg Munshī 2003, 2: 967). Likewise, Mīr Faṭḥ Allāh Shīrāzī (d. 997/1589), the celebrated Safavid polymath and inventor credited with bringing Islamic philosophy to Mughal India, earned this biographical encomium:

He was a very learned scholar and superior to all the scholars of Khurasan, Iraq and India in his mastery of all types of rational and traditional sciences; during his lifetime he had no peer in all the world. He was also skilled in the occult sciences (*ʿulūm-i gharība*), particularly various types of talismans and spellcasting (*ṭilismāt*, *nīranjāt*).

(*Niẓām al-Dīn Aḥmad* 1931–1935, 2: 457)

In view of their elite status, the term used in the earlier passage became standard in Arabic, Persian and Turkish for these sciences as a distinctive unit: *al-ʿulūm al-gharība*, the rare, difficult, paranormal and wondrous sciences, which may be read as a sociological descriptor as much as an epistemological one. Thus only 2% to 15% of scholars captured in Arabic and Persian biographical dictionaries of the 9th/15th to the 13th/19th centuries achieved fame specifically

as occultists – roughly the same percentage, that is, that achieved fame as physicians or *ḥadīth* specialists (Murphy 2010, 90). Occult science, quite simply, had come to be a universal marker of prestige. Alchemy is here an exception, given the threat it represented to economic and hence political stability; a majority of Muslim alchemists labored in anonymity (Harris 2022; Artun 2013). And yet the historical and artifactual record shows that a much broader spectrum of society was involved in medical practices and hadith studies, precisely in view of their prestige. The same principle held true for the occult sciences (including medicine), which were practiced more or less competently by saints and charlatans alike; occult-scientific amateurs abounded. Elite sources (like biographical dictionaries) were thus concerned with *quality control*.

That the Arabic adjective *gharīb* means “elite” and not “strange” in this context is confirmed already in the 6th/12th century by no less a luminary than the philosopher-theologian Fakhr al-Dīn Rāzī (d. 606/1210), who in his *Compendium of the Sciences* (*Jāmi‘ al-‘ulūm*), the first comprehensive Persian encyclopedia of the rational and religious sciences, introduces his eastern scholarly readership to the new occult science of geomancy (*‘ilm al-raml*) by specifically styling it *gharīb*. In his entry on physiognomy (*‘ilm al-firāsa*), an occult science well known from Hellenic antiquity, he further glosses *gharīb* as *sharīf* – precisely meaning “elite” (Fakhr al-Dīn Rāzī 2003, 268, 432; on the many nuances of this descriptor see Zadeh 2020, 2023; on Rāzī as occultist, see Noble 2021). As a trenchant critique of the prevailing Aristotelian–Avicennan classification of all occult sciences as applied natural sciences only, and hence less capable of scientific precision, Rāzī here makes a revolutionary move: he reclassifies certain occult sciences, including judicial astrology (*‘ilm al-aḥkām*), geomancy and jinn magic (*‘ilm al-‘azāyim*), as *mathematical sciences*, for the first time in the Western tradition as a whole. It is in this Rāzian revolution that we can see the seeds of the Occult-Scientific Revolution that was to follow, whereby Aristotle was increasingly if respectfully abandoned in favor of Plato and Pythagoras. In other words: Muslim occult scientists, like their Jewish and Christian peers, found they could best marry Plato to Aristotle by means of Pythagoras, sage, mage and saint in equal measure, on the basis of their personal experience. Significantly for the history of science, the history of religion and the history of empire alike, Pythagoras’s Islamic reincarnation was widely held to be Imam ‘Alī – fountainhead of imperial sacral power (*walāya*) and mathematical occult science for Sunni and Shi‘i alike (Moin 2012; Melvin-Koushki 2018a).

II.9.2 The occult sciences in classifications of the sciences

The epochal process by which the occult sciences became simultaneously natural, mathematical and religious sciences, and hence ubiquitous in early modern Islamicate societies, may best be traced in those Arabic and Persian classifications of the sciences (*taṣnīf al-‘ulūm*) produced between the 4th/10th and the 11th/17th century. In such works, quite simply, the occult sciences (often formally designated as such) constituted roughly *half* of the Islamicate natural and mathematical sciences for a full millennium, right down to, and in some cases through, the colonial rupture (Melvin-Koushki 2017). Naturally, this percentage varies widely from century to century, from region to region and from author to author. But even manifestly anti-occultist encyclopedias were obliged to include many occult sciences as a matter of course; and the most influential classifications of the sciences in the later Arabo-Persian tradition are, without exception, heavily pro-occultist.

We have already seen how Fakhr al-Dīn Rāzī inaugurated this process by reclassifying several occult sciences, most notably astrology, from the natural to the mathematical. And to be a mathematical science, according to early modern Platonic–Aristotelian cosmology, is to have celestial, even spiritual, potency – hence Imam ‘Alī’s new status as a mathematical genius. But even the natural sciences were increasingly valorized in their own right. Here it must be emphasized that

Maslama al-Qurtubī's (d. 353/964) *Goal of the Sage* (*Ghāyat al-ḥakīm*), famed in Latinate Europe as the *Picatrix*, and among the most influential of all grimoires, was sometimes used as a primer in Aristotelian natural (occult) science till at least the 11th/17th century. Indeed, together with equally landmark works on astrology by Abū Ma'shar ([171–272/787–886]; Latin: Albumasar), it was a vector by which Aristotle entered Latin scholarship (Lemay 1962, 1993).

While occult sciences were perennially popular at Islamicate courts from the Abbasids onward, the 6th/12th to 8th/14th centuries, as noted, stand as a major inflection point in the history of Islamicate science and empire in tandem. For it was then that occultism, together with Sufism and Alidism, was made central to the new, experimentalist modes of millennial sovereignty and philosopher-kingship being developed in Ilkhanid Iran and Mamluk Egypt in particular. These modes would be fully realized over the next three centuries in the form of competing Timurid, Aqquyunlu, Safavid, Mughal and Ottoman imperialisms, each pressed into ever more radical avenues of ideological experimentation by their common Turko-Mongol Perso-Islamic rivalry (Melvin-Koushki 2018a). To reiterate: the post-Mongol, plague-wracked, intensely cosmopolitan early modern era was the golden age of Islamicate occultism, and the occult-scientific tenor of early modern Islamicate empire without precedent and parallel in Western intellectual-imperial history (Melvin-Koushki 2018b; Fleischer 2009). That the same era was also the golden age of Judeo-Christian occultism is not an accident. But the latter was decidedly less imperial, and the Christian colonialism of later centuries was violently antimagical (Hanegraaff 2012).

This sea change, this earlier Scientific Revolution, is both heralded by and enshrined in the two most influential classifications of the sciences ever produced in the Islamicate encyclopedic tradition as a whole, Mamluk and Ilkhanid-Injuīd respectively, which served as models for most subsequent Islamicate such works:

1. *Guidance for the Seeker of the Sublimest of Goals* (*Irshād al-qāṣid ilā asnā l-maqāṣid*), in Arabic, by Ibn al-Akfānī (d. 749/1348), a Cairene polymath who perished in the plague; and
2. *Jewels of Sciences Delightful to Behold* (*Nafāyis al-funūn fī 'arāyis al-'uyūn*), in Persian, by Shams al-Dīn Muḥammad Āmulī (d. 752/1352), professor (*mudarris*) at Sultaniyya under the Mongol rulers Öljeitü (r. 704–716/1304–1316) and Abū Sa'īd (r. 716–736/1316–1335), then patron of the Injuīd Shaykh Abū Ishāq (r. 743–54/1343–53) in Shiraz.

Following a general Avicennan schema, the first classifies all occult sciences as derivative natural sciences, a category they dominate. Here is his list of the applied natural sciences in full:

Medicine (*'ilm al-ṭibb*), veterinary medicine and falconry (*'ilm al-bayṭara wa-l-bayzara*), physiognomy (*'ilm al-firāsa*), oneiromancy (*'ilm ta'bīr al-ru'yā*), judicial astrology (*'ilm aḥkām al-nujūm*), magic (*'ilm al-siḥr*), talismans (*'ilm al-ṭilasmāt*), illusionism (*'ilm al-sī-miyā*), alchemy (*'ilm al-kīmīyā*), agriculture (*'ilm al-filāḥa*).

As Ibn al-Akfānī clarifies, these natural sciences concern either simple bodies or composite bodies or both. Thus, astrology concerns celestial simple bodies, and talismans terrestrial or elemental simple bodies. As for composite bodies, illusionism concerns those that have no constitution (*mizāj*), while those that have constitutions are further subdivided into those that do not have souls (alchemy) and those that do have souls. Here there are either insensible souls (agriculture) or sensible (veterinary medicine for animals, medicine for humans). Physiognomy then inquires into the inner states of rational human souls while they are conscious, and oneiromancy does the same while they are unconscious. That science that concerns both simple and composite bodies is magic (Melvin-Koushki 2017, 172–3).

Ibn al-Akfānī posits, in short, an *astrology–talismans–magic* continuum as the very backbone of natural philosophy and psychology alike, running the epistemological–ontological gamut from celestial simple bodies to terrestrial or elemental composite bodies and allowing the competent philosopher–scientist experiential control of the cosmos. Even *agriculture* is here conceptualized as an occult science. On the basis of such evidence, Gutas (2018) suggests that occult science represents the continuation of Avicennan natural science, even though Ibn Sīnā (d. 428/1037) himself, the second Aristotle, was no booster of the occult sciences. That Ibn al-Akfānī could be not only an Avicennan but also such an enthusiastic occultist testifies to the conceptual shift that intervened between the 5th/11th and 8th/14th centuries.

Where this Mamluk offering is highly succinct and dry, however, its Ilkhanid–Injuīd counterpart is expansive, gorgeous and epistemologically revolutionary. Shams al-Dīn Muḥammad Āmulī also employs an Avicennan schema, synthesizing it with the then standard basic division between the Islamic or latter-day sciences (*‘ulūm-i avākhir*) and the philosophical or ancient sciences (*‘ulūm-i avā’l*), and covering an unprecedented number of both. Yet he also seeks to transcend both schemas to move toward a new, more universal conception of knowledge – one in which, crucially, occult sciences like geomancy are to be reclassified as *mathematical*, following Fakhr al-Dīn Rāzī’s radical move of a century and a half before. The various branches of the natural and mathematical sciences are as follows:

Branches of natural science (*furū‘-i ṭabī‘ī*): medicine (*‘ilm-i ṭibb*), alchemy (*‘ilm-i kīmīyā*), letter magic (*sīmīyā*), oneiromancy (*‘ilm-i ta’bīr*), physiognomy (*‘ilm-i firāsāt*), judicial astrology (*‘ilm-i ahkām-i nujūm*), active properties (*‘ilm-i khavāṣṣ*), physiological professions (*‘ilm-i ḥiraf al-ṭabī‘a*) (these including the two divinatory techniques of scapulomancy (*aktāf*) and spasmatomancy (*ikhṭilāj-i a’zā*)), breath control (*‘ilm-i dam*), meditation and visualization (*‘ilm-i vaḥm*) (*dam* and *vaḥm* being Indian yogic sciences).

Branches of mathematical science (*furū‘-i riṣāṣī*): planetary theory (*‘ilm-i hay’at*); optics (*‘ilm-i manāẓir*); the Middle Books (*mutavassitāt*); calculation (*‘ilm-i ḥisāb*); algebra (*‘ilm-i jabr u muqābala*); surveying (*‘ilm-i masāḥa*); constellations (*‘ilm-i zuvar-i kavākib*); star tables, ephemerides and astrolabes and other observational instruments (*‘ilm-i arqām u a’māl-i zīj u taqvim u asṭurlāb u ālāt-i raṣādī*); administrative geography (*‘ilm-i masālik u mamālik*); magic squares (*‘ilm-i vafq-i a’dād*); geomancy (*‘ilm-i raml*); mechanics and wondrous devices (*‘ilm-i ḥiyāl*); games of strategy (*‘ilm-i malā’ib*).

Here, too, natural science is almost entirely occult, but the *Jewels of Sciences* further testifies to the progressive mathematization of the occult sciences, that is, their formal reclassification from the natural (*ṭabī‘ī*) to the mathematical (*riṣāḍī*) sciences. Again, this epochal development, peculiar to the Persianate East, must be considered central to the *mathesis* narrative going forward.

Equally epochally, the *Jewels of Sciences* is also the first encyclopedia to register the contemporary *sanctification* of occult science, and lettrism in particular. Āmulī designates the science of Sufism (*‘ilm-i taṣawwuf*) the supreme Islamic science, equal in importance to all the other religious and traditional sciences combined, and lettrism the supreme Sufi science, that is, *the science of walāya* (Melvin-Koushki 2017, 148–51; on this sanctification process generally see Gardiner 2014; Saif 2017).

Building on this Egyptian–Iranian precedent, the leading occultists of the next century – Timurid and Ottoman scholars radiating from Mamluk Cairo, and calling themselves, significantly, the New Brethren of Purity, after the shadowy 4th/10th-century group of Neopythagorean occultists of the same name – synthesized the parallel processes of mathematization and sanctification to reconceptualize lettrism as the supreme *Alid-Pythagorean universal-imperial* science, a total

physics-metaphysics allowing the scholarly or royal adept comprehensive access to and experiential control over the various levels of being (Melvin-Koushki 2017; Fleischer 2009). Also in Mamluk Cairo, alchemy too was rendered a universal *mathematical-natural-religious* science by Aydemir al-Jildakī (d. c. 743/1342), the most important and prolific Muslim alchemist after (pseudo-)Jābir ibn Ḥayyān; all subsequent Arabic alchemy is Jildakian (Harris 2022; Artun 2013). The mathematization-sanctification of occult science generally and lettrism and alchemy specifically continued apace in the Persian-speaking world over the next centuries. Thus Safavid encyclopedias – even more heavily Neopythagorean-occultist in tenor – seize on these two sciences as both the primary mode of philosophical praxis and the best means of harnessing *walāya* – which is to say, of *magically incarnating the Imams*. It was then that ‘Alī ibn Abī Ṭalīb, inventor of letter divination (*jafr*), was hailed the chief mathematician of Islam (Melvin-Koushki 2017).

For some early modern encyclopedists, moreover, the sanctification of certain natural-mathematical occult sciences was so total as to justify their reclassification exclusively as religious sciences. A case in point is the Ottoman polymath Ṭāshkōprizāde Aḥmed (d. 968/1561), who in his comprehensive *Key to Felicity and Lamp to Mastery* (*Miftāḥ al-sa‘āda wa-miṣbāḥ al-siyāda*), greatly expanding on Ibn al-Akfānī, structures the totality of Islamic scientific and religious knowing according to an explicitly *lettrist* schema, for the first time in the Arabo-Persian tradition. At the same time, he classes lettrism proper not as a mathematical science but as a religious science and specifically as a subset of Quranic exegesis (*tafsīr*); the surest route to the heart of scripture is *magic* (Melvin-Koushki 2016). The Two Books doctrine remains intact; the emphasis, however, is on the primacy of the Quran as model of reality rather than on that model’s call to mathematize the cosmos. Embodying the scientific mainstream, Ṭāshkōprizāde’s offering became the primary model for all subsequent Ottoman encyclopedias, particularly Ḥajjī Khalīfa’s ([d. 1067/1657]; Kātib Chelebi) landmark *Removal of Doubts About the Names of Books and Disciplines* (*Kashf al-zunūn ‘an asānī l-kutub wa-l-funūn*).

This shift of emphasis would seem to reflect a larger Ottoman scholarly tendency to “de-mathematize” astronomy and to return occult sciences like astrology and geomancy to the natural sciences, apparently as a means of differentiating Ottoman learning from its Timurid-*cum*-Safavid model and rival, committed to an equally sacralizing, magical *mathesis* (cf. Fazlhoğlu 2014, 35; Brentjes 2010). For Ṭāshkōprizāde’s Safavid peers went to the opposite extreme: as noted, they singled out lettrism and alchemy as the two ultimate mathematical and Islamic-Imamic, which is to say universal, sciences (Melvin-Koushki 2017). In both Persianate cases, however, the first Sunni and the second Shi‘i, the moral cannot be more explicit – nor more divergent from contemporary developments in Inquisition-Reformation Europe, which was plagued by witch hunts: *Islam is to be lived as Magic*.

While much more research into the Islamicate encyclopedic tradition remains to be done, and the history of the occult sciences in Islam is still mostly unwritten, the overall trend from the medieval to early modern periods is clear. The natural, the mathematical and even the religious sciences became increasingly and disproportionately occult, and the greatest metaphysicians and physicians of early modern Islamdom were often also its greatest mages.

II.9.3 Occult-scientific experimentalism as early modernity

The historical study of Islamicate occult science is thus an especially efficient means of exploding colonialist-orientalist myths as to the eternal unmodernity of Islam: it is a project that cuts to the quick of what it means to be modern – and scientific.

For Muslim occult scientists constantly testify to the *empirical* nature of their operations. That is to say, Islamicate early modernity, like its Christianate parallel, is characterized by an

increasingly heavy emphasis on the experimentally verifiable nature of science, occult and otherwise. Here the central Arabic terms are *taḥqīq*, “independent verification,” and *tajriba*, “experiential essaying”; the latter’s semantic evolution closely parallels that of its Latin cognate *experientia* (Janssens 2004; Langermann 2014; Melvin-Koushki 2018b; cf. Gutas 2018; Brentjes 2018[CB], 10, 74–7). This, it must be emphasized, represents a significant departure from Aristotelian precedent, wherein *empeiria* is a mode of general knowledge considerably inferior to *epistēmē*, “science.” The empirical nature of occult science was already asserted by such authorities as Abū Maʿshar, systematizer of conjunction astrology, who influentially declared his discipline to be a science founded on analogical reasoning (*qiyās*) and experience (*tajriba*), the latter entailing careful observation (*muʿāyana*) and direct discovery (*al-wujūd al-ḥādir*; Burnett 2002, 210, 212). An identical argument is made for medicine (often classified as an occult science) by Ibn Hindū (d. 409/1019 or 420/1029), who rather uses the terms *raṣad* and *mushāhada* for observation and personal inspection respectively (Gutas 2002, 279–80). And Ibn Sīnā himself asserted *tajriba* to be superior to Aristotelian inductive reasoning (McGinnis 2010, 47–52; Gutas 2012).

By the early modern period, such categorizations became standard for the occult sciences as an epistemological unit. Geomancers, lettrists, astrologers and alchemists now insisted without exception on the *mujarrab* nature – the *reproducibility* – of their findings and procedures, now laid equal claim to *taḥqīq*. Representing the state of the art at the turn of the 10th/16th century, for example, ʿAlī Ṣafī (d. 939/1533), son of the great Timurid preacher, polymath and occultist Ḥusayn Vāʾiḡ Kāshifī (d. 910/1505), asserts in his programmatic lettrist manual *Amulet of Protection from the Trials of Time* (*Ḥirz al-amān min fitan al-zamān*): “Experiential essaying aimed at manifesting the effects and properties of the letters has often been successfully repeated; and experience is among the forms of proof that produce certainty (*tajriba yakī az isbāt-i yaqīn ast*)” (Melvin-Koushki 2017, 157 no. 103). And as we saw earlier, in a major departure from medieval precedent, the same period saw an explosion in the production of Arabic and Persian occult-scientific manuals, imperial grimoires, that explicitly invite the reader to master their contents independently of a teacher (cf. El-Rouayheb 2015[CB]). Personal experience of the seen and the unseen alike became an imperative.

Crucially for the history of science, such an occultist-experientialist ethos is highly reminiscent of Baconian experimentalism, that other basis of scientific modernity – wherein natural-mathematical science was just as frequently occult. Francis Bacon (d. 1626), the patron saint of Royal Society experimentalism, was himself a strong advocate of divinatory sciences like geomancy, although he did not go so far as to classify them as mathematical like his Muslim peers (Langman 2010). In this Persianate context, it is historically ironic that Bacon pursued the rehabilitation of what he imagined to be ancient *Persian* magic, as well as alchemy, astrology and various forms of divination, as the core of his new explicitly Protestant form of natural philosophy. He invoked magic not as superstition but as superstition’s cure (Josephson-Storm 2017, 44–51).

In sum, as in Latinate Europe, Persianate occult science was widely and emphatically embraced as a means of demonstrating the validity of Neoplatonic and especially Neopythagorean physics-metaphysics, and hence a primary driver of *mathesis*. But unlike in Europe, it also became foundational to early modern Islamic, Mongol, Persian and Alexandrian-universalist *imperialism*. That is to say, Bacon’s famous dictum – *knowledge is power* – would seem to have been taken more seriously, and to much greater political effect, by early modern Muslim ruling elites and the occultists that served them. Contemporary Christian imperial ideologies could hardly scientifically or materially compete (Melvin-Koushki 2018c). While it should be noted that Bacon’s famous phrase has been routinely misunderstood – in context, it means rather “[God’s] knowledge is itself power” (Josephson-Storm 2017, 46–7) – it accords neatly with the primary objective of our early modern Muslim occult philosophers: the attainment of *theosis*,

or *theomimesis*, which drove in turn their projects to help imperially realize the Pythagorean One in time and space.

II.9.4 Conclusion

The Western cosmology now regnant, materialist though it may be, still descends in some part from that established first in the Islamicate world and then paralleled in the Christianate world during the early modern period, whereby the cosmos is to be scientifically understood, and imperially and technologically controlled, as a mathematical text. For what are Google and the Utah Data Center but imperial Comprehensive Prognosticons? And even Ronald Reagan (r. 1981–1989) had his court astrologer. So while the various occult sciences are no longer patronized by state and society to the great degree they once were, and we now believe Modernity to be the antithesis of Magic (Hanegraaff 2012), the Occult-Scientific Revolution achieved by our Muslim, Jewish and Christian forebears continues to inform our cosmology and hence our practice of science and empire.

Is occult science Science, then, in early modern Islamicate societies specifically and Western societies generally? Yes, naturally – and mathematically. We cannot historically explain *mathesis*, a project associated with Pythagoras from late antiquity to the present, or the rise of early modern experimentalism, the very matrix of scientific modernity, without it. That Magic is often simultaneously Religion – which is to say, concerned as much with mind as with matter – as well as Empire – which is to say, concerned as much with power as with knowledge – does not exclude it from history of science. Likewise, that Islam as imagined and lived was (and still is) often tantamount to Magic does not exclude it either from Western civilization. As a host of early modern Muslim scholarly and ruling elites declared, following Aristotle’s advice to Alexander: *Islam is magic; magic is science; science is empire*.

Note

- 1 Consolidated bibliography.

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